Mingyu Kang

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Professional Summary

Quantum physicist with experience spanning three major layers of the quantum computing stack: hardware, error correction, and applications. Research conducted at the theory-experiment interface – connecting atomic physics, quantum control, quantum error correction, and quantum chemistry – across academia and industry. Developed open-source quantum software with >55 GitHub stars.

Education

Duke University 2020 – 2025
Ph.D. in Physics *Durham, NC*

- Dissertation: *Towards Quantum Advantage with Trapped Ions*

Stanford University 2014 – 2020
B.S. Physics, with Distinction and Departmental Honors · GPA: 4.02/4.3 *Stanford, CA*

Experience

Postdoctoral Scholar Aug. 2025 – Present
UC Berkeley, Dept. of Chemistry · Supervisor: Prof. K. Birgitta Whaley

- Performing resource estimation for early-fault-tolerant quantum chemistry algorithms, targeting practical quantum advantage benchmarks

Intern Jun. 2024 – Sep. 2024
QuEra Computing

- Modeled two-qubit gate error sources in neutral-atom processors using detailed atomic physics and system noise models, providing quantitative guidance for hardware improvement

Intern (virtual) Mar. 2021 – Sep. 2021
IonQ Inc.

- Developed efficient protocols for high-precision motional-mode characterization in trapped-ion quantum computers, reducing calibration overhead (Quantum Science and Technology, 2023; patent pending)

Ph.D. Researcher Aug. 2020 – May 2025
Duke University, Dept. of Physics & Duke Quantum Center · Advisor: Prof. Kenneth R. Brown

- Designed pulse-optimization algorithms that improved two-qubit gate fidelities on trapped-ion processors; results experimentally validated (Physical Review Applied, 2021 & 2023; patent granted)
- Developed erasure-conversion quantum error correction scheme exploiting trapped-ion error structure, achieving significantly higher fault-tolerance thresholds (PRX Quantum, 2023)
- Co-designed and led trapped-ion analog quantum simulation experiments of molecular energy transfer dynamics (Nature Reviews Chemistry, 2024; Nature Communications, 2025)

Undergraduate Researcher Jan. 2019 – Jun. 2020
Stanford University · Advisor: Prof. Amir Safavi-Naeini

- Applied optimal control to maximize Bell-state fidelity in a nanomechanical quantum memory coupled to a superconducting qubit

Sergeant & Signal Analyst Sep. 2016 – Jun. 2018
ROK Army SEC Security Lab

- Built GUI application for encoding, decoding, and identifying convolutional error-correcting codes of various types

Open-Source Projects

QUITS: A modular Qldpc code circUIT Simulator | github.com/mkangquantum/quits 2025

- Led development of a modular, flexible circuit-level simulator for quantum low-density parity check (LDPC) codes; resulted in a peer-reviewed publication (Quantum, 2025); earned >55 GitHub stars

Awards & Grants

NSF Quantum Leap Challenge Institute Seed Grant 2023 – 2024

Simulating Dissipative Electron Transfer with Trapped-Ion Systems

- $100,000 research funding; collaboration with four other PhD students

John T. Chambers Fellowship, Duke Fitzpatrick Institute for Photonics 2020 – 2021

$10,000; one of four awardees at Duke University

Kwanjeong Scholarship, Kwanjeong Educational Foundation 2014–2016, 2018–2020

$55,000/yr; one of seven undergraduate awardees in South Korea

Publications

Published Journal Articles

- **MK***, Y. Zhang*, K. R. Brown, and T. Barthel, *Non-Gaussian Phase Transition and Cascade of Instabilities in the Dissipative Quantum Rabi Model*, Phys. Rev. A **113**, L061703 (2026). arXiv:2507.07092
- H. Ren, Y. Zhang, W. M. Billings, R. Tomann, N. V. Tkachenko, **MK**, M. Head-Gordon, and K. B. Whaley, *Error-mitigated nonorthogonal quantum eigensolver via shadow tomography*, Phys. Rev. Res. **8**, 013268 (2026). arXiv:2504.16008
- **MK***, Y. Lin*, H. Yao, M. Gökduman, A. Meinking, and K. R. Brown, *QUITS: A modular Qldpc code circUIT Simulator*, Quantum **9**, 1931 (2025). arXiv:2504.02673
- K. Sun*, **MK***, H. Nuomin, G. Schwartz, D. N. Beratan, K. R. Brown, and J. Kim, *Quantum Simulation of Spin-Boson Models with Structured Bath*, Nat. Commun. **16**, 4042 (2025). arXiv:2405.14624
- **MK**, H. Nuomin, S. N. Chowdhury, J. L. Yuly, K. Sun, J. Whitlow, J. Valdiviezo, Z. Zhang, P. Zhang, D. N. Beratan, and K. R. Brown, *Seeking a Quantum Advantage with Trapped-Ion Quantum Simulations of Condensed-Phase Chemical Dynamics*, Nat. Rev. Chem. **8**, 340–358 (2024). arXiv:2305.03156
- Q. Liang, **MK**, M. Li, and Y. Nam, *Pulse Optimization for High-Precision Motional-Mode Characterization in Trapped-Ion Quantum Computers*, Quantum Sci. Technol. **9**, 035007 (2024). arXiv:2307.15841
- **MK**, W. C. Campbell, and K. R. Brown, *Quantum Error Correction with Metastable States of Trapped Ions using Erasure Conversion*, PRX Quantum **4**, 020358 (2023). arXiv:2210.15024
- K. Sun, C. Fang, **MK**, Z. Zhang, P. Zhang, D. N. Beratan, K. R. Brown, and J. Kim, *Quantum Simulation of Polarized Light-Induced Electron Transfer with a Trapped-Ion Qutrit System*, J. Phys. Chem. Lett. **14**, 6071–6077 (2023). arXiv:2304.12247
- Z. Jia, S. Huang, **MK**, K. Sun, R. F. Spivey, J. Kim, and K. R. Brown, *Angle-Robust Two-Qubit Gates in a Linear Ion Crystal*, Phys. Rev. A **107**, 032617 (2023). arXiv:2210.04814
- **MK**, Q. Liang, M. Li, and Y. Nam, *Efficient Motional-Mode Characterization for High-Fidelity Trapped-Ion Quantum Computing*, Quantum Sci. Technol. **8**, 024002 (2023). arXiv:2206.04212
- **MK**, Y. Wang, C. Fang, B. Zhang, O. Khosravani, J. Kim, and K. R. Brown, *Designing Filter Functions of Frequency-Modulated Pulses for High-Fidelity Two-Qubit Gates in Ion Chains*, Phys. Rev. Applied **19**, 014014 (2023). arXiv:2206.10850
- **MK**, Q. Liang, B. Zhang, S. Huang, Y. Wang, C. Fang, J. Kim, and K. R. Brown, *Batch Optimization of Frequency-Modulated Pulses for Robust Two-Qubit Gates in Ion Chains*, Phys. Rev. Applied **16**, 024039 (2021). arXiv:2104.06887

Service & Outreach

Editorial Board, *Physical Review A* Jan. 2026 – Dec. 2027

Early Career member

Journal Referee

Entropy, Nature Communications, npj Quantum Information, Physical Review A, Physical Review Letters, Physical Review Research, Physical Review X, PRX Quantum, Physica Scripta, Quantum, Science Advances

Co-host, YouTube series *Quantum News Monthly* Sep. 2023 – Sep. 2024

- Reviewed selected papers published each month in the field of quantum physics and computing